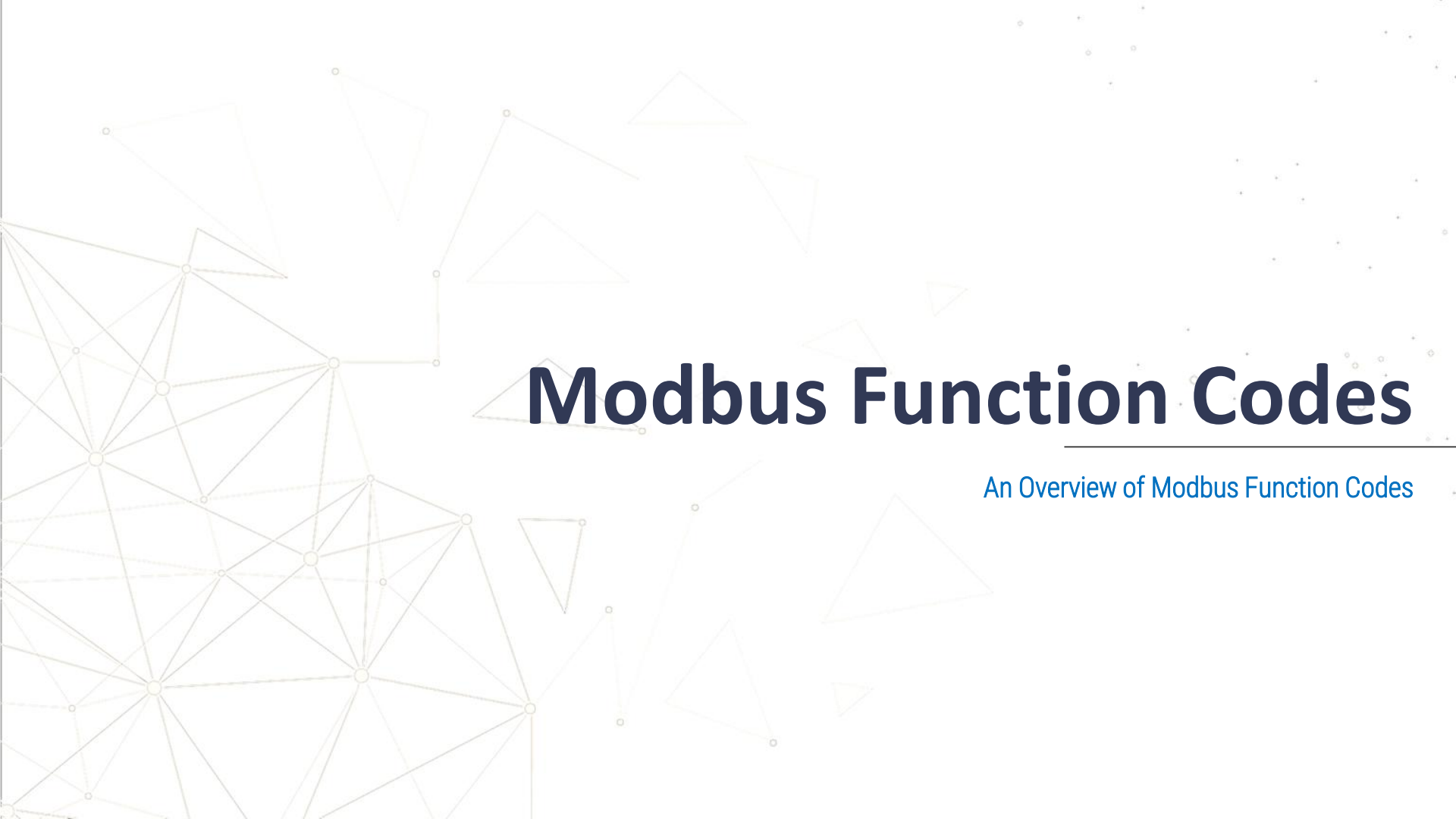


The background features a complex network of thin, light-colored lines and dots, forming a web-like structure. Scattered throughout are various triangles of different sizes and orientations, some with solid outlines and others with dashed or dotted lines. The overall aesthetic is technical and modern, with a color palette dominated by light blues, greys, and off-whites.

Modbus Function Codes

An Overview of Modbus Function Codes

Modbus Function Codes

- Modbus Function Codes Introduction
- Common Modbus Function Codes
- Requests and Responses

Modbus Function Codes Introduction

Introduction to Modbus Function Codes

- **Function Codes:** Commands that a Modbus master uses to instruct a slave device to perform specific actions, such as reading or writing data. They are an essential part of the communication between devices in a Modbus network.

Slave Address	FC	Register Address	Number of Registers to Read	CRC
07	03	00 0D	00 01	2 Bytes



Function Code for Reading Holding Register

Common Modbus Function Codes

Read Coils (01): Reads the status of 1-bit coils.

Read Discrete Inputs (02): Reads 1-bit discrete inputs.

Read Holding Registers (03): Reads multiple 16-bit holding registers.

Read Input Registers (04): Reads multiple 16-bit input registers.

Write Single Coil (05): Sets the value of a single coil (0 or 1).

Write Single Register (06): Writes a value to a single 16-bit holding register.

Write Multiple Coils (15): Writes values to multiple coils.

Write Multiple Registers (16): Writes values to multiple holding registers.

Function Type		Function Name	Function Code
Bit Access	Discrete Inputs	Read Discrete Inputs	2
	Coils	Read Coils	1
		Write Single Coil	5
		Write Multiple Coils	15
16-Bit Access	Physical Input Registers	Read Input Registers	4
	Internal Registers or Physical Output Registers	Read Multiple Holding Registers	3
		Write Single Holding Register	6
		Write Multiple Holding Registers	16

Requests and Responses

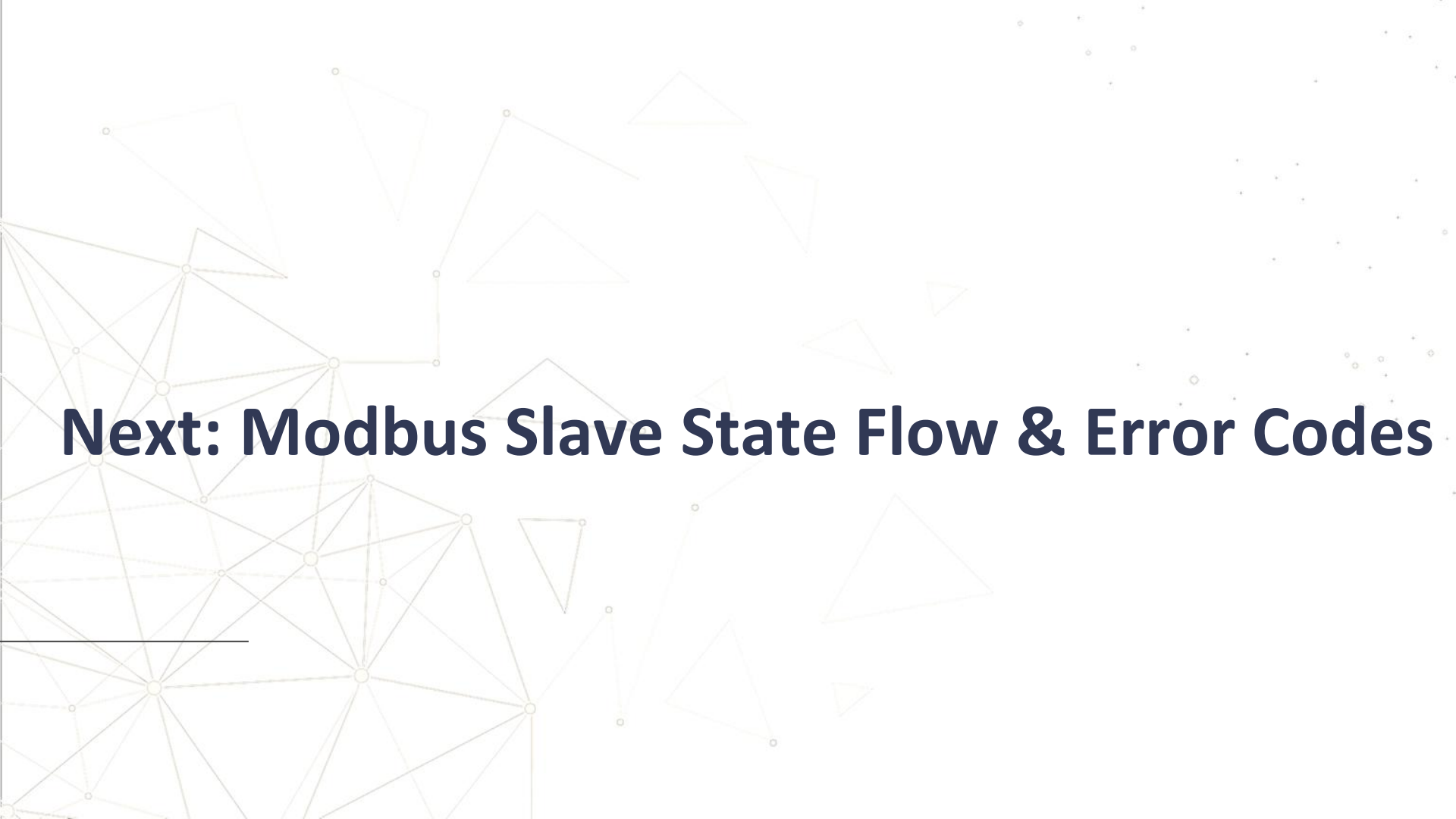
- **Request:** Sent by the master, containing the function code addresses, and data.

Slave Address	FC	Register Address	Number of Registers to Read	CRC
07	04	00 02	00 01	2 Bytes

- **Response:** Sent by the slave, echoing the function code and returning data or acknowledgment.

Slave Address	FC	Byte Count	Data	CRC
07	04	02	00 64	2 Bytes

- **Normal Response:** Echoes the function code (e.g., 04 for reading input registers) and provides the requested data.
- **Error Response (Exception):** Modifies the function code by adding 0x80 and includes an exception code to indicate the error (e.g., 0x84 for an error in reading input registers).

The background features a complex network of thin, light-colored lines and small circles, forming a web-like structure. Overlaid on this are several translucent triangles of various sizes and orientations, some of which are filled with a light yellow or green color. The overall aesthetic is technical and modern.

Next: Modbus Slave State Flow & Error Codes